Chapter 1: Coupled Numerical Experiments

In this chapter, we couple the locally mass-conserving MFEM and FVM methods to perform numerical simulations of a partially molten mantle, with the remaining uncertainty in porosity ϕ addressed through phase package evaluation. We describe the implementation details and coupling strategy that integrate the MFEM, FVM, and the eutectic phase package. Before tackling the fully coupled problem, we first examine an alternative porosity evolution scenario based on a prescribed melting rate. We then present an example that combines the MFEM, FVM, and eutectic phase package into a fully coupled computation. The discussion restricts to pseudo-1D scenarios and subsequently extends to full 2D simulations.

1.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ denote the cell-averaged solution to the transport system (**??**) at time t^n. Similarly, let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ represent the solution to the Darcy-Stokes system at the same time t^n. Finally, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n obtained from the transport result $\mathcal{U}^n$.

For each time step the time t^n, we begin by reconstructing the dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ using ML-WENO reconstruction at each data point of interest. The eutectic phase package then provides the corresponding phase parameters $\mathcal{W}^n$ based on these point wise values. With $\mathcal{W}^n$ determined in particular with ϕ^n, we solve the Darcy-Stokes system at the fixed time t^n to obtain $\mathcal{V}^n$. Finally, the cell-averaged solution $\mathcal{U}^{n+1}$ at the next time t^{n+1} is then obtained by forward marching of the transport system. The complete procedure is summarized as a coupled solver in Algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let $\mathcal{U}^0$ be given as initial data.

1: **for** Time level $n = 0, 1, 2, \ldots, n_{\max} - 1$ **do**

2: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.

3: Using ϕ^n in $\mathcal{W}^n$, solve the Darcy-Stokes system (**??**)–(**??**) for $\mathcal{V}^n$.

4: Given $\mathcal{W}^n$ and $\mathcal{V}^n$, compute dimensionless effective velocity (**??**) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi \check{\mathbf{v}}_l + D_i(1-\phi)\check{\mathbf{v}}_s}{\phi + (1-\phi)D_i}, \quad \check{\bar{\mathbf{v}}} = \phi \check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

5: March forward the transport system (**??**) to the next time for $\mathcal{U}^{n+1}$.

6: **end for**

1.1.1 Discontinuous Porosity

Due to the presence of no melt regions and distinct eutectic phase regions, the reconstructed values of the dimensionless enthalpy and mass concentration may differ on the two sides of an edge shared by adjacent elements. This, in turn, can lead to two different porosity values being computed for the same edge. However, in a conforming MFEM scheme, it is essential to assign a single consistent and physically appropriate porosity value on each element edge.

Therefore, we evaluate the harmonic average of the two values of porosity ϕ_e^+ and ϕ_e^- at each quadrature points on both sides of an element edge e to obtain

$$\overline{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}. \tag{1.1}$$

The harmonically averaged porosity $\overline{\phi}_e$ is then used in the continuous compaction equation (**??**), where the face integral is converted to an edge integral via the divergence theorem.

Importantly, when the porosity on one side of the edge vanishes, the harmonic average also becomes zero. This property enforces the physical condition that no liquid can pass directly through an edge adjacent to a no-melt region.

1.2 Pseudo One-Dimensional Column Tests

In this section, we test the coupled algorithm on a two-dimensional $M \times N$ grid with a pseudo one-dimensional compaction column problem. We study a compacting column aligned along the y-direction, defined over the domain $y \in [-H, 0]$ and $x \in [-L, L]$. Two scenarios are investigated.

- Prescribed melting case: The MFEM and FVM solvers are coupled without phase computation. Instead, a simple prescribed melting function $\Gamma(\mathbf{x})$ is used.
- Phase calculation case: Phase transition and porosity evolution are computed simultaneously, providing a complete coupling between the MFEM, FVM, and eutectic phase package.

For the two tests presented below, we fix the domain dimension in the x-direction $L = 0.1$. We fix mesh resolution in the x-direction as $M = 2$ and pick different resolutions in the y-direction. The domain dimension and mesh resolution in the y-direction are chosen accordingly. We fix the parameter $\Theta = 0$ as well. The code is implemented in PETSc/C++, with implementation details given in the appendix.

1.2.1 Prescribed Melting Cases

Before coupling the mechanical, transport and phase packages together into one integrated computation, we first illustrate a simpler model that couples MFEM and FVM solvers without involving the phase package. In this reduced setting, the porosity distribution is not determined by phase-splitting calculations. Instead, we prescribe the melting rate or, equivalently, the porosity evolution through a fixed source function distributed in space as $\Gamma(\mathbf{x})$.

We express conservation of liquid mass by introducing the prescribed melting term $\Gamma(\mathbf{x})$ as a source term on the right-hand side of the advection equation as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \tag{1.2}$$

where the diffusion term is being dropped for simplicity in this case.

We prescribe the following boundary conditions for the MFEM and the FVM solvers, respectively, to replicate the behavior of the one-dimensional problem within the two-dimensional grid.

Consider first the Darcy-Stokes MFEM solver. On the bottom edge $y = -H$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with an essential boundary conditions with a constant upwelling velocity v, so

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e v\, ds;$$

- Tangential Stokes velocity is prescribed with an essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with an essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

On the top edge $y = 0$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with natural boundary conditions, i.e., traction free,

$$t_0 = 0;$$

- Tangential Stokes velocity is prescribed with a zero essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with a natural boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

On the left and right Edges $x = \pm L$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0, \ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} \, ds = 0;$$

- Tangential Stokes velocity is prescribed with the natural boundary condition of free traction,

$$t_0 = 0;$$

- Normal Darcy flux is prescribed with a non-permeable essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

Finally, for the transport FVM solver, on the bottom edge $y = -H$, impose that:

- A corresponding Dirichlet boundary condition is prescribed on the inflow boundary aligning with the essential boundary condition specified in the MFEM solver.

On the top edge $y = 0$, impose that:

- A free outflow boundary condition is imposed, consistent with the free traction boundary condition applied in the MFEM solver.

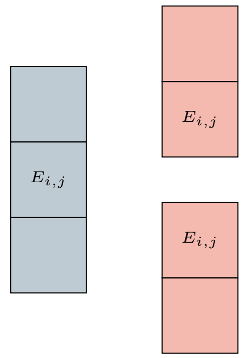


Figure 1.1: An exhibition of a set of 1D ML-WENO (3,2) stencils on a rectagular mesh.

On the left and right edges $x = \pm L$, impose that:

- A no-flow boundary condition is prescribed consistent with the zero-flow Dirichlet boundary used in the MFEM solver.

For simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 1.1, since higher-order accuracy in the x-direction is unnecessary for this problem. We use the second order SSP Runge-Kutta time integration scheme defined in Equation (**??**).

We consider two different scenarios, each characterized by distinct prescribed melting functions and initial conditions:

Case 1: A piece-wise constant source function.

$$\Gamma_1(y) = \begin{cases} 2.5^{-3}, & -1.5 < y \leq 0, \\ 0, & -2.0 \leq y \leq -1.5. \end{cases} \tag{1.3}$$

We assign an initial condition of no melting everywhere as

$$\phi_1(y, 0) = 0.0, \quad y \in [-2.0, 0.0]. \tag{1.4}$$

Figure 1.2 illustrates the temporal evolution of porosity, Stokes and Darcy velocities, Stokes and Darcy pressures, as well as the effective pressure, defined as $p_{\text{eff}} = p_s - p_l$.

The third panel, corresponding to dimensionless time $\check{t} = 1325$, demonstrates that the system has reached a steady state after sufficient evolution.

Case 2: We next consider a slightly more complicated scenario in which a melting source is trapped beneath the bottom of an existing porosity channel. The trapped melting source produces a step in porosity, thereby induces an instantaneous perturbation in the compaction rate. The source function is

$$\Gamma_2(y) = \begin{cases} 0, & -1.5 < y \leq 0, \\ 1.5 \times 10^{-3}, & -1.8 < y \leq -1.5, \\ 0, & -2.0 \leq y \leq -1.8, \end{cases} \tag{1.5}$$

where we assign an initial condition with an existing porosity channel as

$$\phi_2(y, 0) = \begin{cases} 5 \times 10^{-2}, & -1.5 < y \leq 0, \\ 0, & -2.0 \leq y \leq -1.5. \end{cases} \tag{1.6}$$

Figures 1.3, 1.4 and 1.5 illustrate the temporal evolution of porosity, Stokes and Darcy velocities and Stokes and Darcy pressures. A propagating discontinuity in porosity forms when the porosity profile decreases upward (see Katz (2022)). Eventually, the system reaches a steady state, characterized by a narrower porosity channel. The simulation demonstrates robust and stable evolution, despite the presence of a highly complicated porosity profile. The solution preserves its overall structure under mesh refinement.

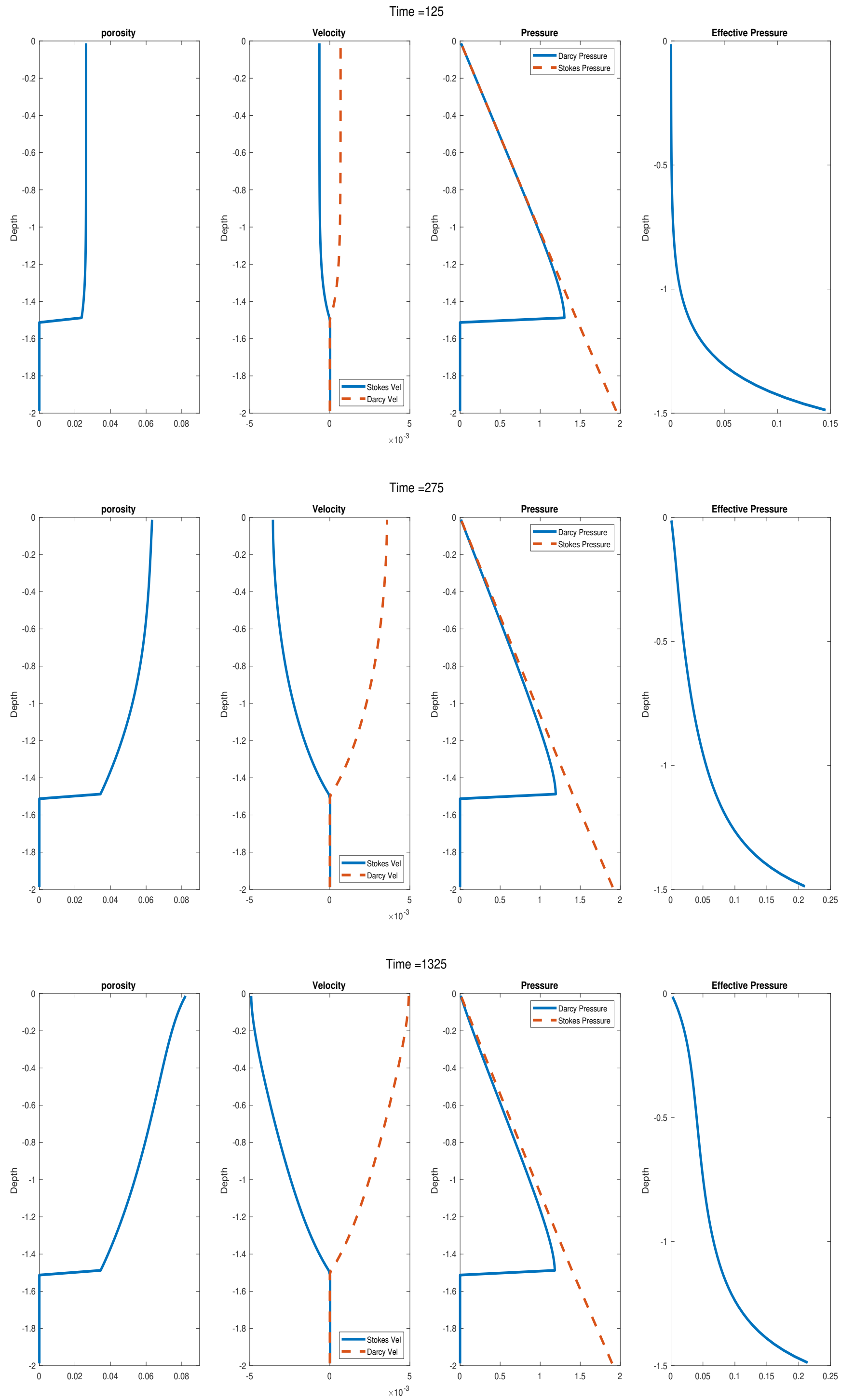


Figure 1.2: Prescribed melting case with constant melting source term 1.3. Shown at time = 125, 275 and 1325, where the system reaches steady state at the time = 1325.

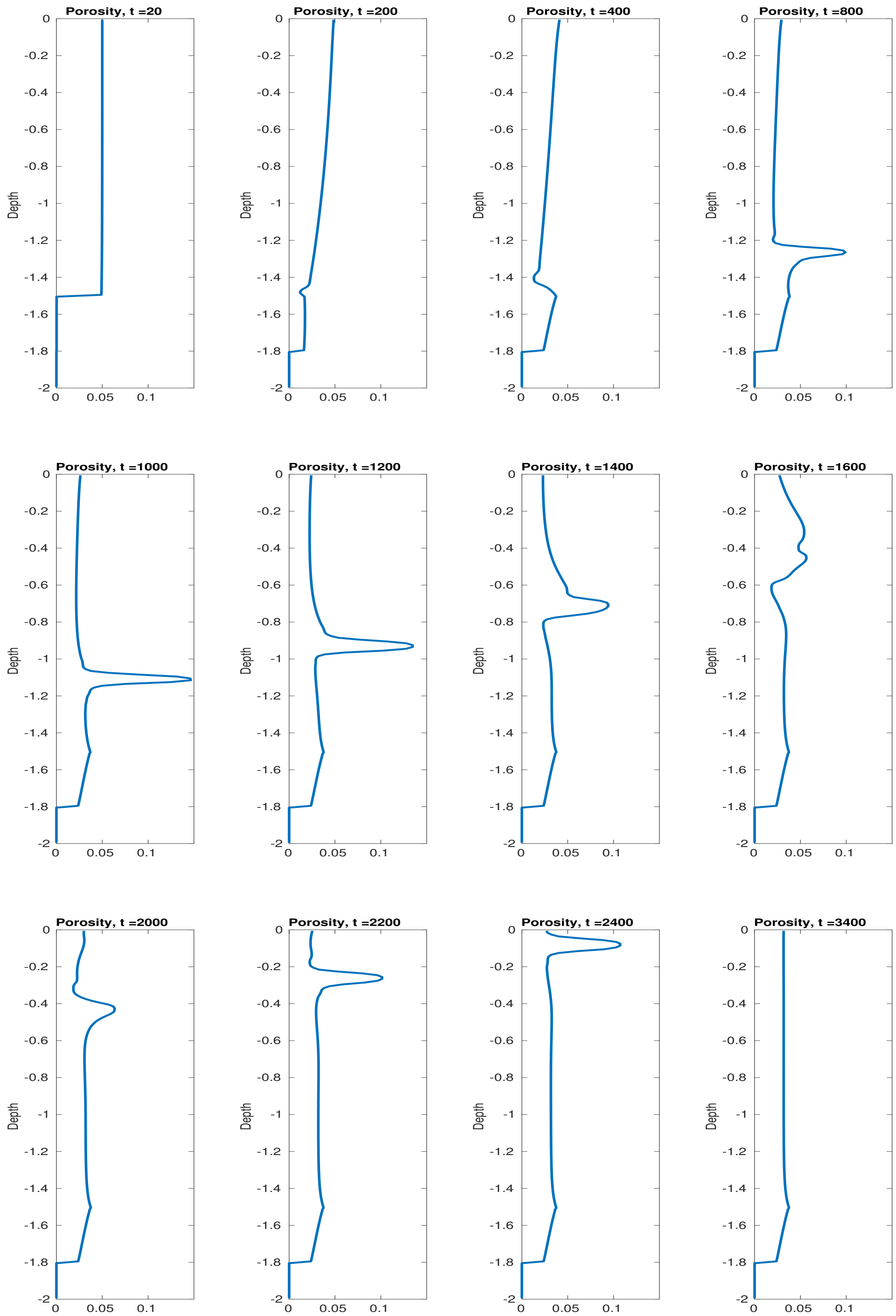


Figure 1.3: Prescribed melting case with trapped melting source term 1.5. Temporal evolution of porosity shown at different time steps.

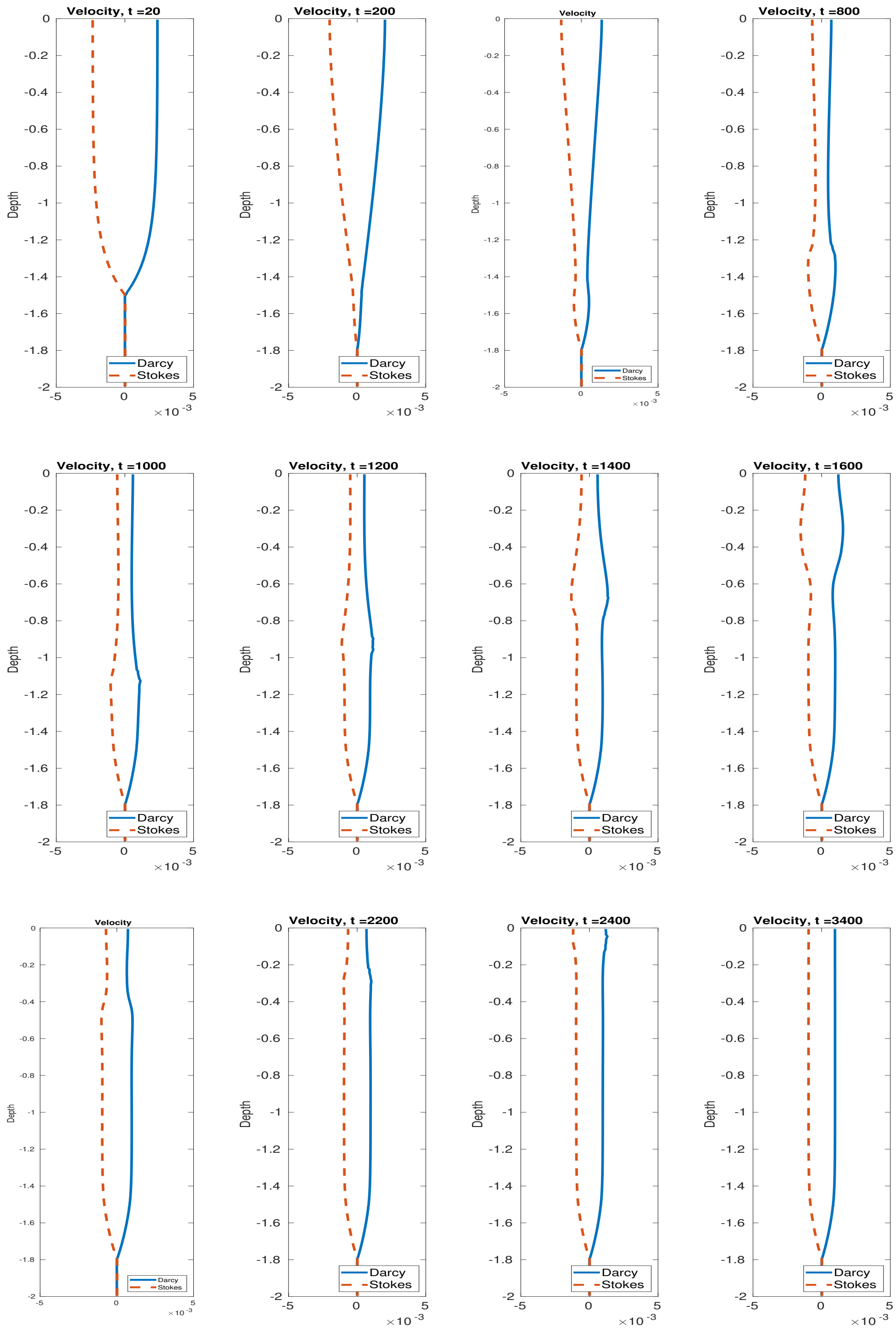


Figure 1.4: Prescribed melting case with trapped melting source term 1.5. Temporal evolution of velocities shown at different time steps.

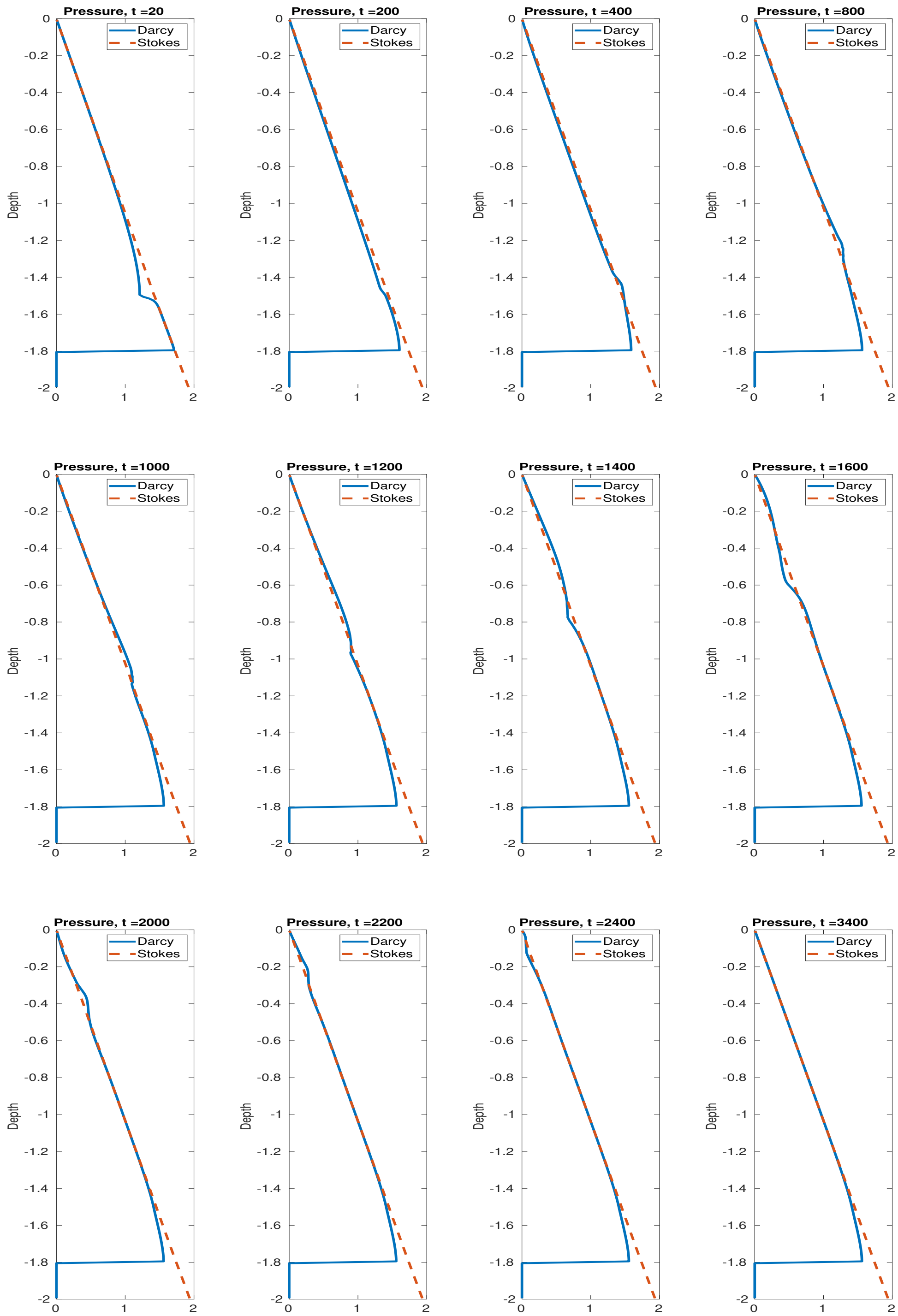


Figure 1.5: Prescribed melting case with trapped melting source term 1.5. Temporal evolution of pressures shown at different time steps.

1.2.2 Phase Coupled Calculation

In this section, we test the MFEM and FVM solvers combined with the eutectic phase package, which gives the fully integrated simulation framework. In this setting, porosity is no longer prescribed through an externally imposed melting function but is instead computed consistently from the transported nondimensional bulk composition and enthalpy.

The computational domain represents an upwelling mantle column, with corner flow neglected near the surface under the simplifying assumption that the column continues its vertical ascent through the lithosphere. This configuration constitutes a minimal yet dynamically representative test case for assessing the stability and accuracy of the coupled algorithm in the presence of phase transitions and melt–solid interactions.

We propose some simplifications for our simulations. First of all, we reconsider the transport equations (**??**) and the simplified version is

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla \cdot \check{\mathbf{v}}_{\mathrm{eff}} C &= 0, \\ \frac{\partial H}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}} \check{T}) - \check{L} \nabla \cdot ((1-\phi)\check{\mathbf{v}}_s) - \frac{\alpha_\rho l_0}{c_p} \check{T} \check{\mathbf{v}} \cdot \mathbf{g} &= 0. \end{aligned} \tag{1.7}$$

In this formulation, the diffusive contributions in both the mass composition and enthalpy equations are neglected. Specifically, the diffusion term $\frac{\phi \mathcal{D}_l}{u_0 l_0} \nabla c_l$ in the liquid phase is omitted due to the fact that the porosity ϕ remains small ($\phi \ll 1$) and the associated diffusivity coefficient $\mathcal{D}_l$ is comparatively negligible compared to effective advective velocities and thus can be neglected and thus can be neglected (See Spiegelman (1996)).

Likewise, while thermal conduction is an important mechanism in mantle convection over geological timescales, its influence on the present simulations is negligible. Two considerations motivate the simplification of dropping the thermal conduction from the enthalpy equation (**??**). First, the temperature profile maintains either a nearly constant or linear variation in space, such that the Laplacian of temperature vanishes or remains small throughout the evolution. Second, we do not impose a non-permeable solidification boundary at the surface, which would otherwise establish sharp thermal gradients and necessitate conductive heat transport across a boundary layer, in which case the diffusivity would no longer be negligible (see Hewitt and Fowler (2008)).

Additionally, given that the primary focus of this study is the phenomenon of partial melting under small-porosity conditions, we adopt the lithostatic pressure as the reference pressure in determining the local melting point (see Katz (2008)). This approximation is justified because the dynamic pressure perturbations associated with porous flow are small compared to the background lithostatic gradient. As the mantle column ascends from depth, the concomitant reduction in lithostatic pressure leads to a decrease in the solidus temperature, thereby inducing some degree of melting in proportion to the degree of decompression.

1.2.2.1 Boundary conditions

In this set of simulations, we do not attempt to prescribe fully physically consistent boundary conditions, owing to the inherent complexity of capturing both the mechanical and thermochemical constraints at the mantle–lithosphere interface. For example, Hewitt and Fowler (2008) formulated boundary conditions that simultaneously satisfy conservation laws and thermodynamic equilibrium at the surface. These constraints introduce additional complexity to the model and the numerical scheme that is not central to the phenomena under investigation here.

By prescribing essential boundary conditions on the bottom and the lateral edges and a natural (free-traction) boundary condition at the top for the Darcy–Stokes MFEM solver, the simulations allow for mechanically reasonable mantle upwelling without artificial confinement. Similarly, the Dirichlet inflow and free outflow conditions for the transport solver ensure a constant supply of chemical species and energy while permitting realistic advective transport, without imposing restrictive thermodynamic constraints. This approach maintains the internal consistency of the coupled MFEM–FVM solver and captures the key dynamics of melt migration, porosity evolution and phase transitions, while reducing computational complexity.

1.2.2.2 Compaction Length

The compaction length is typically defined as

$$\delta_c = \sqrt{\frac{\kappa_\phi(\zeta + 4\eta/3)}{\mu_l}}, \tag{1.8}$$

where ζ and η are bulk and shear viscosities of the solid mantle, respectively. The compaction length is an important intrinsic length scale that characterize the interaction between the pressure difference and variations in fluid flux (see Mckenzie (1984) and Spiegelman (1993)).

In this work, we pick a simple constitutive relationship as shown in equation (**??**), which is $\zeta = \mu_s/\phi$ in the notation of this section and a power-law permeability as $\kappa_\phi = \kappa_0\phi^2$. Assuming that the shear viscosity η is much smaller than the bulk viscosity ζ, the appropriately scaled compaction length can be written in terms of the nondimensionalization length scale as

$$\delta_c = \sqrt{\frac{\kappa_0\mu_s}{\mu_l}\phi} = l_0\sqrt{\phi}, \tag{1.9}$$

where l_0 represents the characteristic length scale used in the nondimensionalization. Thus $\delta_c = \delta(\phi)$ depends on the (evolving) porosity. In geological settings, a constant reference porosity is typically prescribed to define the compaction length. However, this assumption is not adopted in the present study, hence we use dimensionless length in the subsequent plots.

1.2.2.3 Numerical Flux

We modify the semi-discretized formulation of a general conservation law, as presented in equation (1.10), to incorporate a "source" term S. The resulting expression can be written as

$$\frac{\partial \overline{u}_E}{\partial t} + \sum_{i=0}^{4} \frac{|e_i|}{|E|} \sum_{g=0}^{G} \omega_{i,g} F(\mathcal{R}_{i,g}^+, \mathcal{R}_{i,g}^-) + \sum_{h=0}^{H} \xi_h S(\mathcal{R}_h) = 0, \tag{1.10}$$

where $\omega_{i,g}$ represents the weight associated to the g-th quadrature point on edge e_i and ξ_h represents the weight associated to the h-th quadrature point on the cell. In practice, we employ the midpoint rule for the numerical integration over cells. This allows us to approximate the face contributions using the cell-averaged transport variables while retaining second-order accuracy.

The adiabatic cooling term $\frac{\alpha_\rho l_0}{c_p} \check{T} \check{\overline{\mathbf{v}}} \cdot \mathbf{g}$ in equation (1.7) is treated as a "source" term, which is straightforward to implement within the semi-discretized formulation. The corresponding Lax-Friedrichs numerical fluxes of the mass composition and the enthalpy across element edges are expressed in terms of the ML-WENO reconstructed values $\mathcal{R}_C$ and $\mathcal{R}_H$ as

$$\begin{aligned} F_C(\mathcal{R}_C^+, \mathcal{R}_C^-) &= \frac{1}{2} \check{\mathbf{v}}_{\text{eff}} \cdot \boldsymbol{\nu} \Big[\mathcal{R}_C^+ + \mathcal{R}_C^- - \alpha_C \big(\mathcal{R}_C^+ - \mathcal{R}_C^- \big) \Big], \\ F_H(\mathcal{R}_H^+, \mathcal{R}_H^-) &= \frac{1}{2} \check{\overline{\mathbf{v}}} \cdot \boldsymbol{\nu} \Big[\check{T}^+ + \check{T}^- - \alpha_T \big(\check{T}^+ - \check{T}^- \big) \Big] - \check{L}(1-\phi) \check{\mathbf{v}}_s \cdot \boldsymbol{\nu}, \end{aligned} \tag{1.11}$$

where $\boldsymbol{\nu}$ is the ourward unit normal of the edge, and $\check{\mathbf{v}}_{\text{eff}}$ and $\check{\overline{\mathbf{v}}}$ denote the effective velocity and phase averaged velocity continuous across the edges. The terms $\check{T}^+ = \check{T}(\mathcal{R}_H^+, \mathcal{R}_C^+)$ and $\check{T}^- = \check{T}(\mathcal{R}_H^+, \mathcal{R}_C^+)$ are temperatures evaluated on both sides of the edge using the eutectic phase package, based on the reconstructed pointwise mass composition and enthalpy. The stabilizers α_C and α_T use the local Lax-Friedrich scheme as shown in equation (**??**), ensuring stability of the flux discretization.

The calculation of the numerical flux (1.11) of the mass composition is not entirely straightforward. The primary transport variables—mass composition and enthalpy—may undergo distinct reconstructions across an edge due to discontinuities and the presence of multiple phase regions. As a consequence, the porosity ϕ can take different values, denoted $\phi^+ = \phi(\mathcal{R}_H^+, \mathcal{R}_C^-)$ and $\phi^- = \phi(\mathcal{R}_H^-, \mathcal{R}_C^-)$. To ensure compatibility in the MFEM discretization, we adopt the harmonic average of the two values, as discussed previously in Section 1.1.1, and defined the compatible porosity on the edge as

$$\overline{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}.$$

In contrast, the bulk mass composition represents a relatively smooth advected quantity. A harmonic average would add artificial damping and suppress mixing, while an arithmetic average is more suitable without over-penalizing discontinuities. Therefore, we adopt

$$c^e = (c^+ + c^-)/2. \tag{1.12}$$

We now revisit the definition of effective velocity defined in equation (**??**) with these interface definitions as

$$\mathbf{v}_{\text{eff}} = \frac{c_l^e\overline{\phi}\mathbf{v}_l + c_s^e(1-\overline{\phi})\mathbf{v}_s}{c_l^e\overline{\phi} + c_s^e(1-\overline{\phi})}, \tag{1.13}$$

where $\overline{\phi}$ is the harmonic-averaged porosity and c_s^e, c_l^e are the arithmetic averaged mass compositions.

Similarly, the phase averaged velocity required in the enthalpy flux is expressed without ambiguity as

$$\check{\mathbf{v}} = \overline{\phi}\mathbf{v}_l + (1-\overline{\phi})\mathbf{v}_s, \tag{1.14}$$

where the same harmonic averaged porosity is employed. This guarantees that in the limiting case $\phi^+ = 0$ or $\phi^- = 0$, the phase-averaged velocity reduces to the solid velocity, thereby preserving the correct physical behavior and preventing artificial transport of enthalpy associated with liquid in no-melt regions.

1.2.2.4 Test results

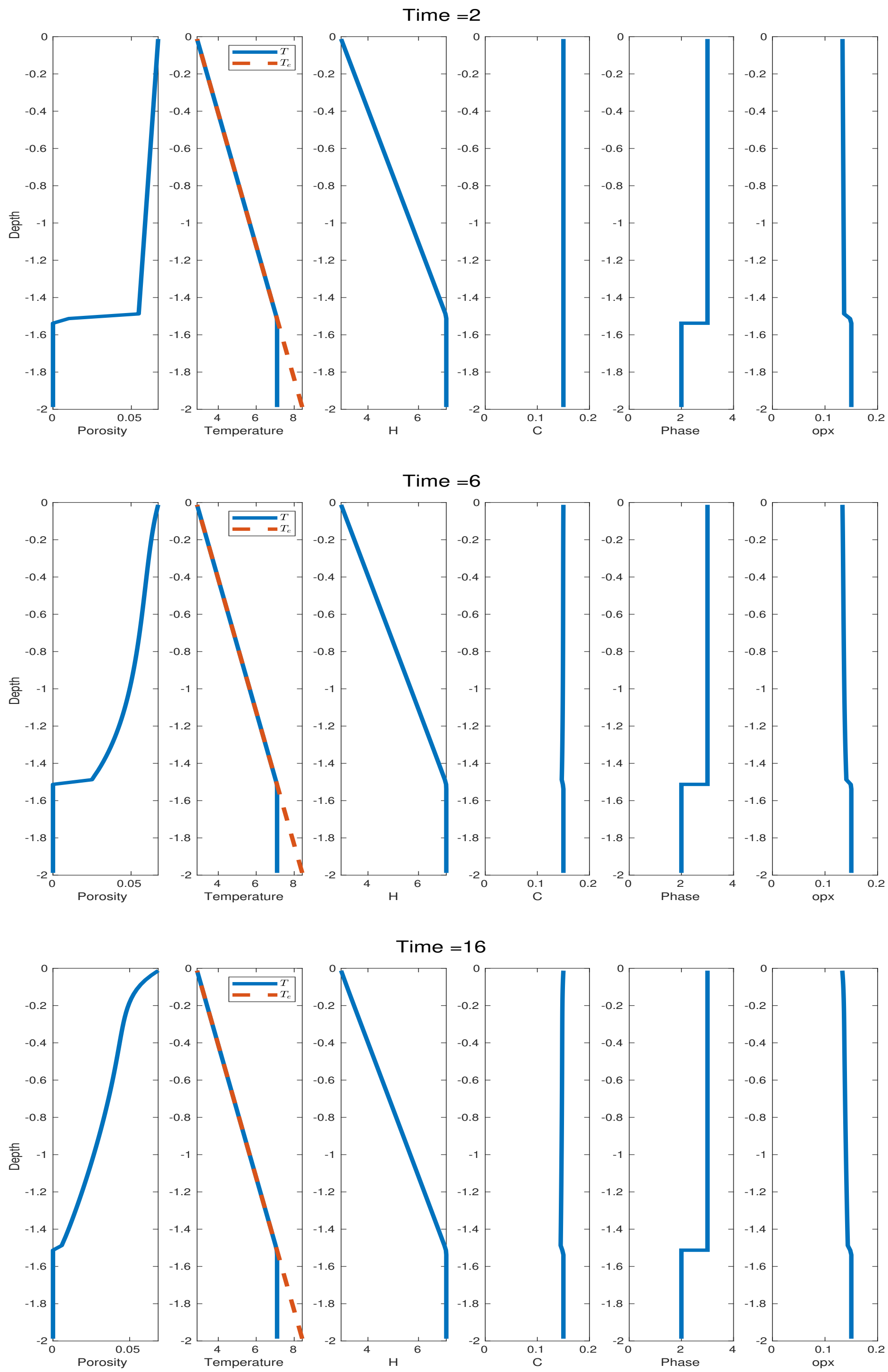
Again, for simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 1.1 for both the mass composition and the enthalpy. We use second order SSP Runge-Kutta time integration scheme defined in Equation (**??**) as well.

Full coupled column testing case $H = 2$: The domain in the y-direction spans two dimensionless lengths $H = 2$ with mesh resolution $N = 80$. We assign an initial condition of the bulk dimensionless mass composition and the bulk dimensionless enthalpy as

$$
\begin{aligned}
H &= \begin{cases} 2.9 - 2.8y, & -1.5 \leq y < 0, \\ 2.9 + 2.8 \times 1.5, & -2 \leq y < -1.5, \end{cases} \\
C &= 0.15.
\end{aligned} \tag{1.15}
$$

Figure 1.6 illustrates the temporal evolution of the primary transport variables, bulk dimensionless mass composition and bulk dimensionless enthalpy, along with the corresponding phase calculation determined from the phase package. From time $t = 2$ to time $t = 6$ we observe a transition from phase region 3 (eutectic melting) to phase region 2 (solid, no melting phase) near the cut $y = -1.5$. The porosity field appears to be highly sensitive to these changes in bulk enthalpy and mass composition, leading to the development of a porosity wave near the upper boundary. This wave-shaped pattern arises from the accumulation of porosity near the top boundary, where the local depletion of mass composition occurs more rapidly than it can be replenished from below. This in turn reduces porosity locally and blocks the draining of liquid further down. This wave-shaped pattern near the top boundary subsequently amplifies and ultimately results in an unstable outcome in the numerical solution around time $t = 130$. The wave-shaped pattern persists under mesh refinement, indicating that it reflects a physical feature of the model, which uses a nonphysical BC, rather than a numerical artifact.

Figure 1.7 shows three characteristic velocities defined on the element edges: the effective velocity transporting chemical species, the phase averaged velocity governing the advection of enthalpy, and the solid velocity carrying latent heat. The result confirms that the local mass conservation is well-preserved as evidenced by the phase averaged velocity remaining constant under the Boussinesq approximation.


Time =2
Time =6
Time =16
Depth
Porosity
Temperature
H
C
Phase
opx
T
T_e

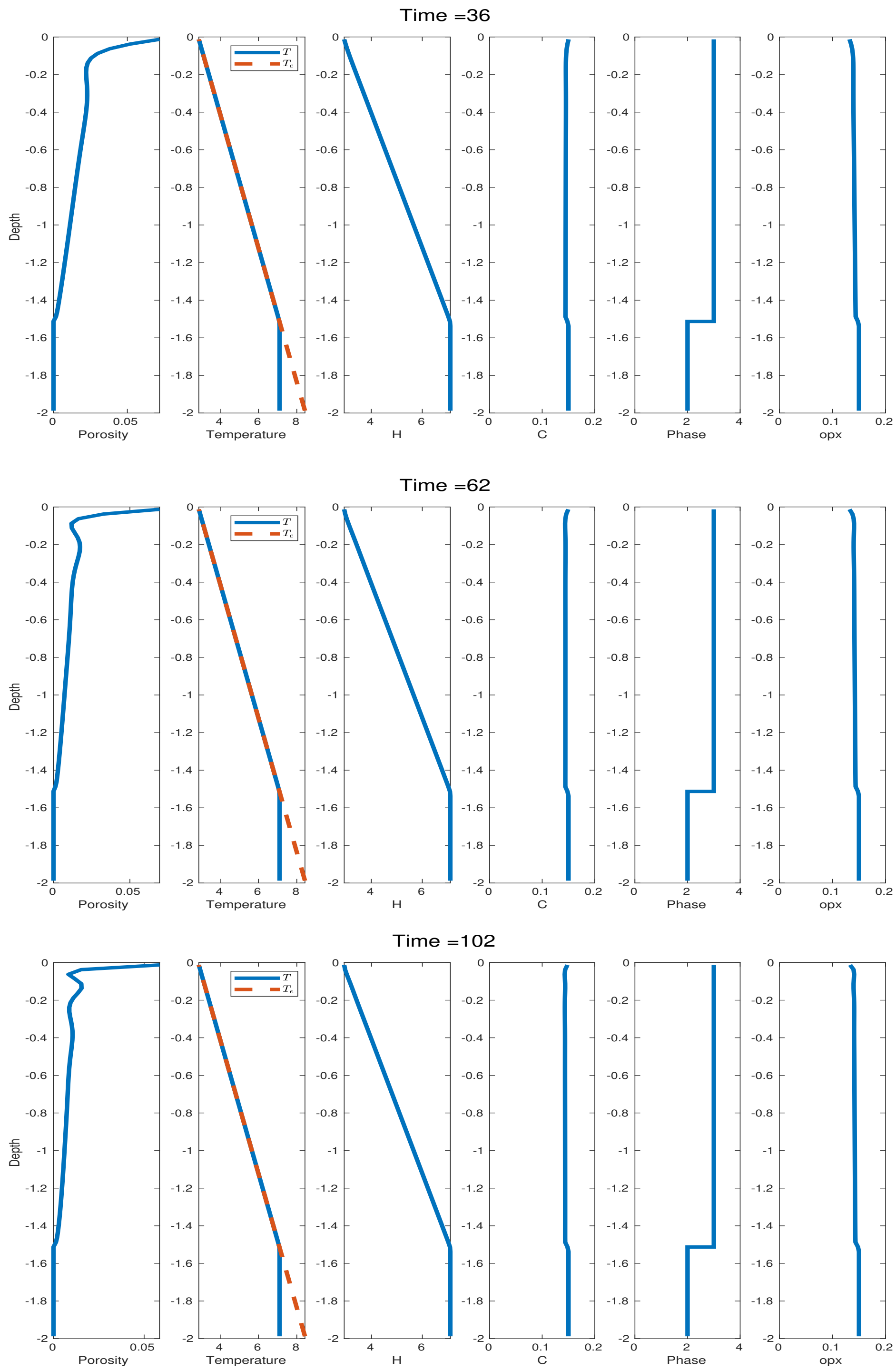
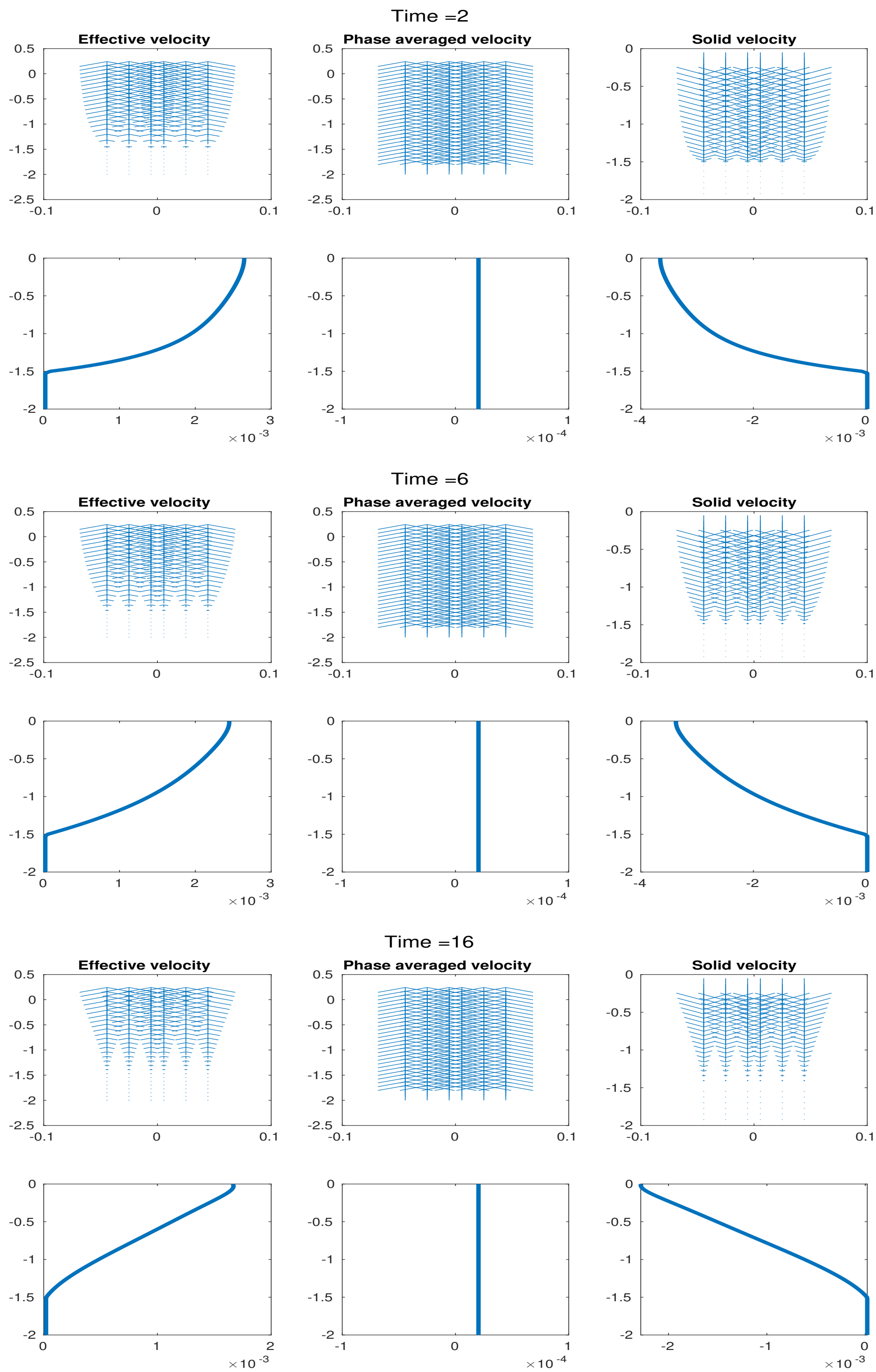


Figure 1.6: Full coupled column testing case $H = 2$. Shown at time = 2, 6, 16, 36, 62 and 102. We present porosity, temperature, dimensionless enthalpy, dimensionless mass composition, phase regions and Opx percentage from left to right.


Time =2
Effective velocity
Phase averaged velocity
Solid velocity
$\times 10^{-3}$
$\times 10^{-4}$
$\times 10^{-3}$
Time =6
Effective velocity
Phase averaged velocity
Solid velocity
$\times 10^{-3}$
$\times 10^{-4}$
$\times 10^{-3}$
Time =16
Effective velocity
Phase averaged velocity
Solid velocity
$\times 10^{-3}$
$\times 10^{-4}$
$\times 10^{-3}$

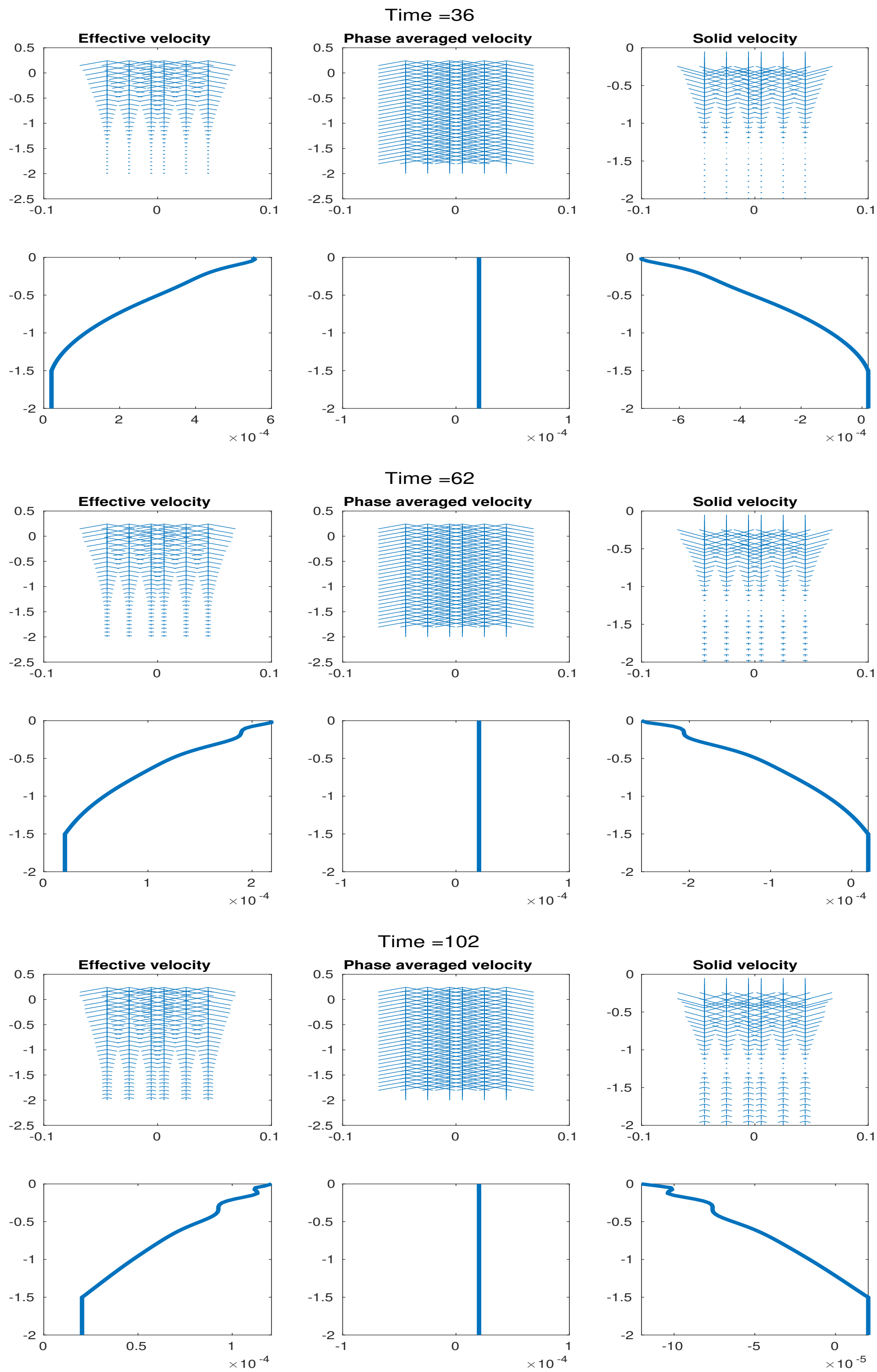


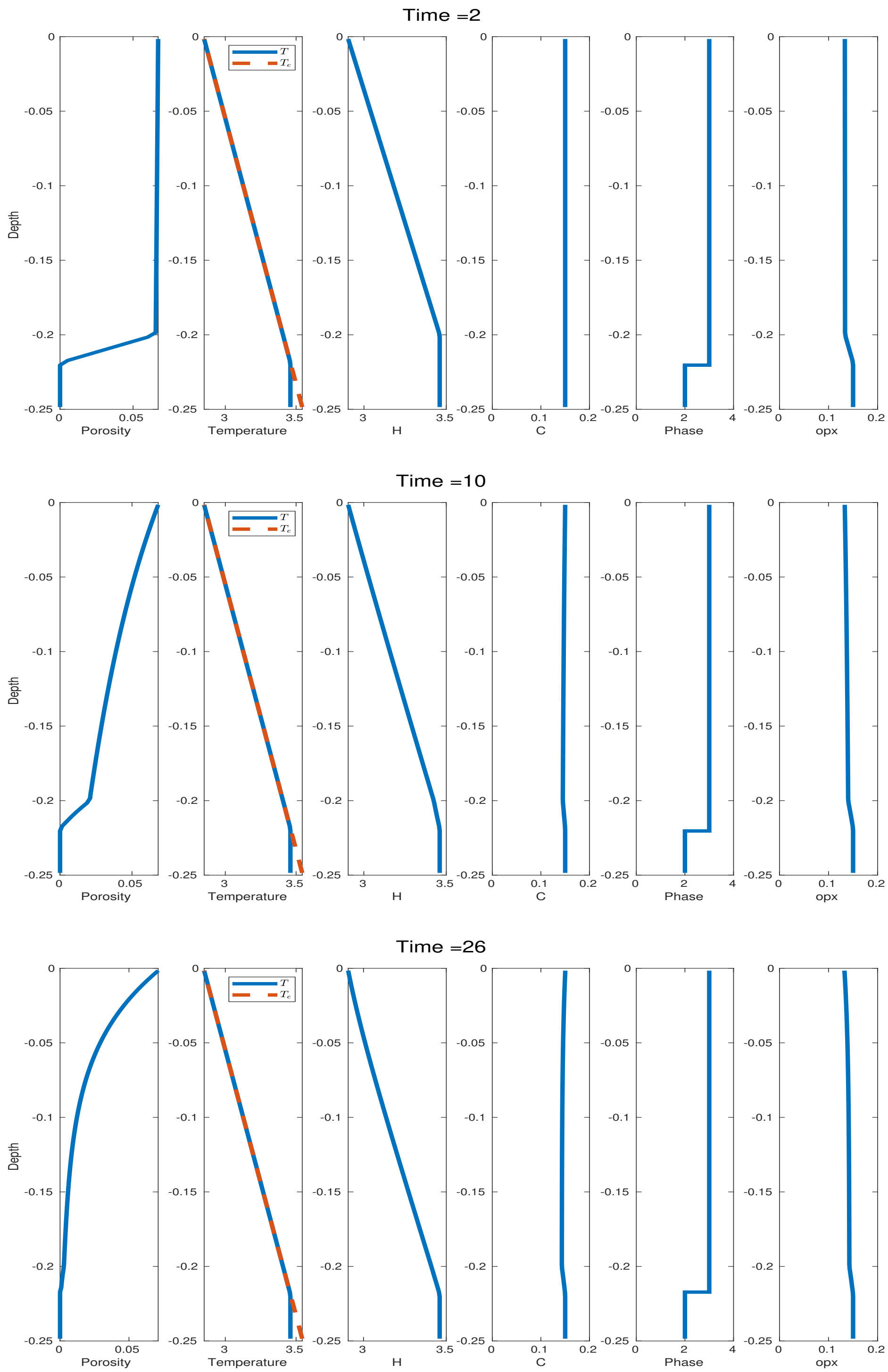
Figure 1.7: Full coupled column testing case $H = 2$. Shown at time = 2, 6, 16, 36, 62 and 102. We present effective velocity, phase averaged velocity and solid velocity from left to right.

Full coupled column testing case $H = 0.25$: The domain in the y-direction spans 0.25 dimensionless lengths $H = 0.25$ with mesh resolution $N = 80$. We assign the same initial condition of the bulk dimensionless mass composition and a similar initial condition of the bulk dimensionless enthalpy with different cutoff as

$$\begin{aligned} H &= \begin{cases} 2.9 - 2.8y, & -0.2 \leq y < 0, \\ 2.9 + 2.8 \times 0.2, & -0.25 \leq y < -0.2, \end{cases} \\ C &= 0.15. \end{aligned} \tag{1.16}$$

Figure 1.8 illustrates the temporal evolution of the primary transport variables, bulk dimensionless mass composition and bulk dimensionless enthalpy, along with the corresponding phase calculation determined from the phase package. Between time $t = 10$, 26 and 40, the system undergoes successive transitions: initially from Phase Region 3 to Phase Region 2, and subsequently returning to Phase Region 3. These transitions occur while maintaining numerical stability throughout the simulation, illustrating the ability of our framework to handle transition between melting and no melting zones. The system reaches steady state at time $t = 240$, while a notable accumulation of porosity remains concentrated near the top boundary. The instability encountered in the previous larger domain does not arise in this case, as the overall porosity is smaller.

Figure 1.9 again presents the three characteristic velocities defined on the element edges: the effective velocity transporting chemical species, the phase averaged velocity governing the advection of enthalpy and the solid velocity carrying latent heat. As the simulation progresses, the effective velocity exhibits a marked increase near the top boundary, consistent with the porosity accumulation observed in Figure 1.8.


Time =2
Time =10
Time =26
Depth
Porosity
Temperature
H
C
Phase
opx
T
T_e

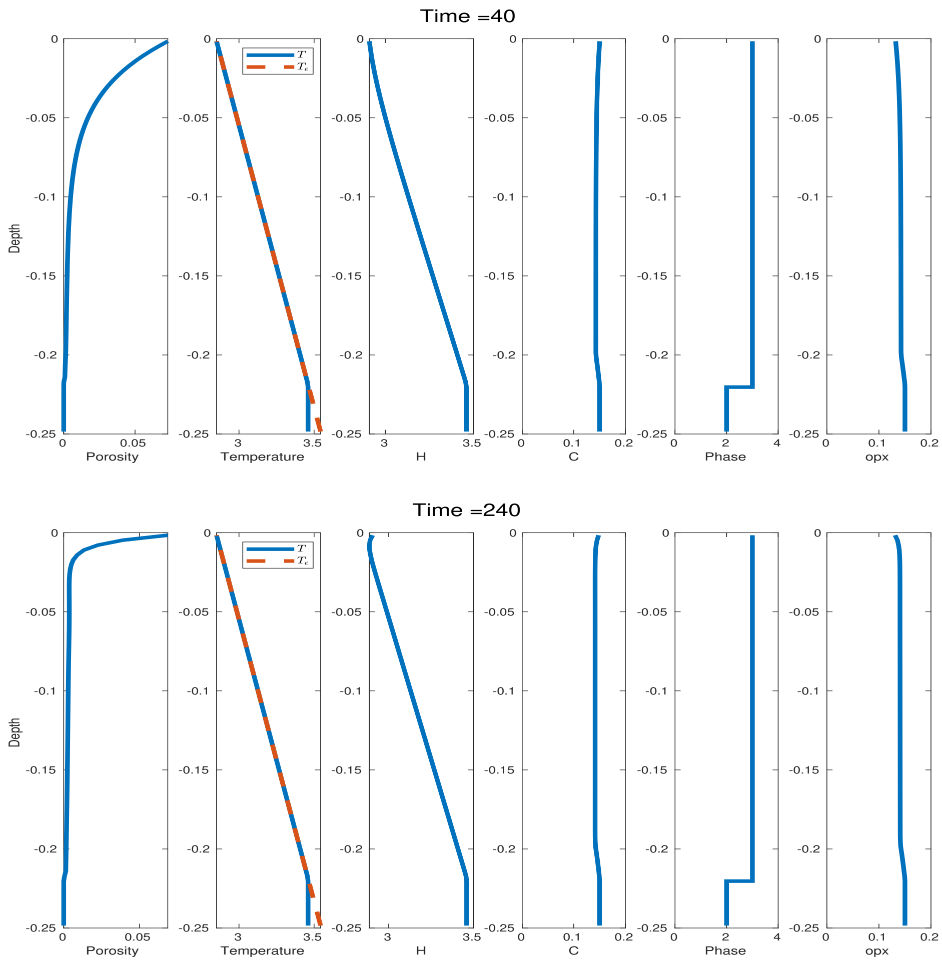
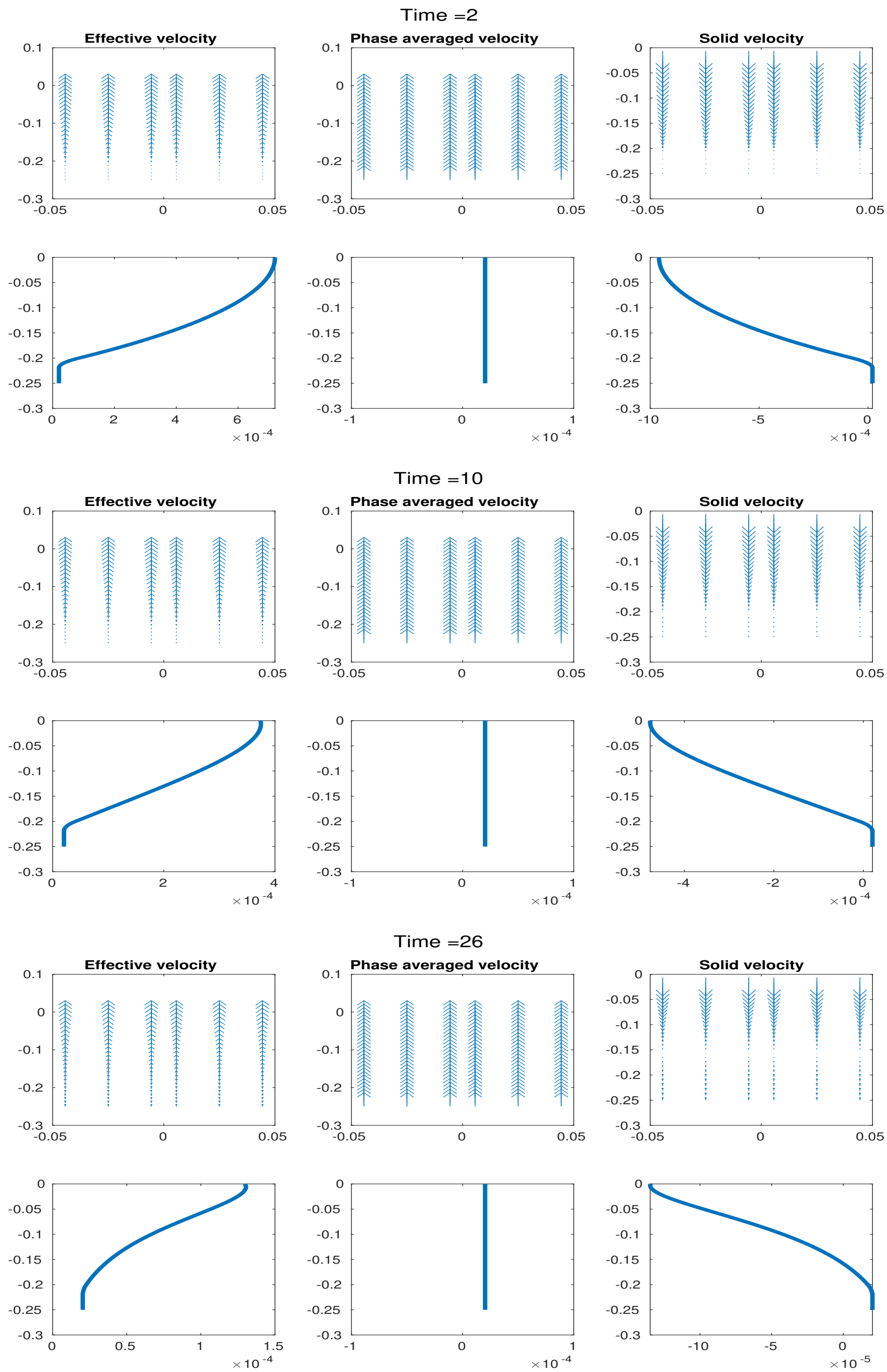


Figure 1.8: Full coupled column testing case $H = 0.25$. Shown at time = 2, 10, 26, 40, and 240. We present porosity, temperature, dimensionless enthalpy, dimensionless mass composition, phase regions and Opx percentage from left to right.



Time =2
Effective velocity
Phase averaged velocity
Solid velocity
Time =10
Effective velocity
Phase averaged velocity
Solid velocity
Time =26
Effective velocity
Phase averaged velocity
Solid velocity

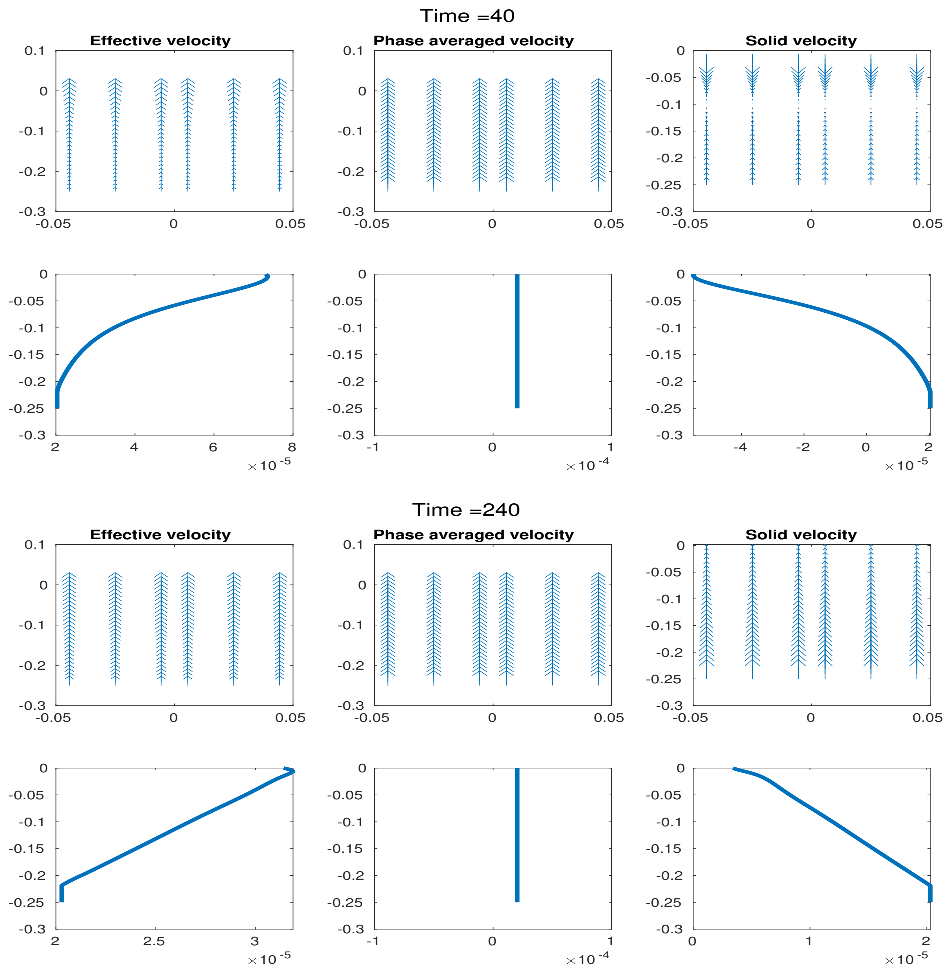


Figure 1.9: Full coupled column testing case $H = 0.25$. Shown at time $= 2$, 10, 26, 40, and 240. We present effective velocity, phase averaged velocity and solid velocity from left to right.

1.3 Two-Dimensional Test

In this section, we test the fully coupled algorithm on a two-dimensional $M \times N$ grid. The computational domain is defined with dimensions $L = 2$ in the x-direction and $H = 2$ in the y-direction. We reuse the boundary conditions from the pseudo–one-dimensional case described in Section 1.2.2: outflow is permitted only through the top boundary, a constant upwelling velocity is imposed at the bottom boundary, and no-flow conditions are applied on the lateral boundaries. The two-dimensional ML-WENO(3,2) stencils, illustrated in Figure **??**, are employed without including the constant stencil. We assign a similar initial condition of the bulk dimensionless mass composition and a similar initial condition of the bulk dimensionless enthalpy as

$$\begin{aligned} H &= \begin{cases} 2.9 - 2.8y, & -1.5 \le y < 0, \\ 2.9 + 2.8 \times 1.5, & -2 \le y < -1.5, \end{cases} \\ C &= 0.1. \end{aligned} \tag{1.17}$$

This numerical experiment is designed to demonstrate the capability of our solver to handle a fully two-dimensional configuration, although the setup itself is not physically meaningful.

Figure 1.10 illustrates a simple two-dimensional simulation incorporating phase calculations. As porosity is drained, the system undergoes stable phase transitions from phase 2 (eutectic melting) to phase 1 (no melting) and from phase 2 (eutectic melting) to phase 3 (super-eutectic melting), without exhibiting numerical instability.

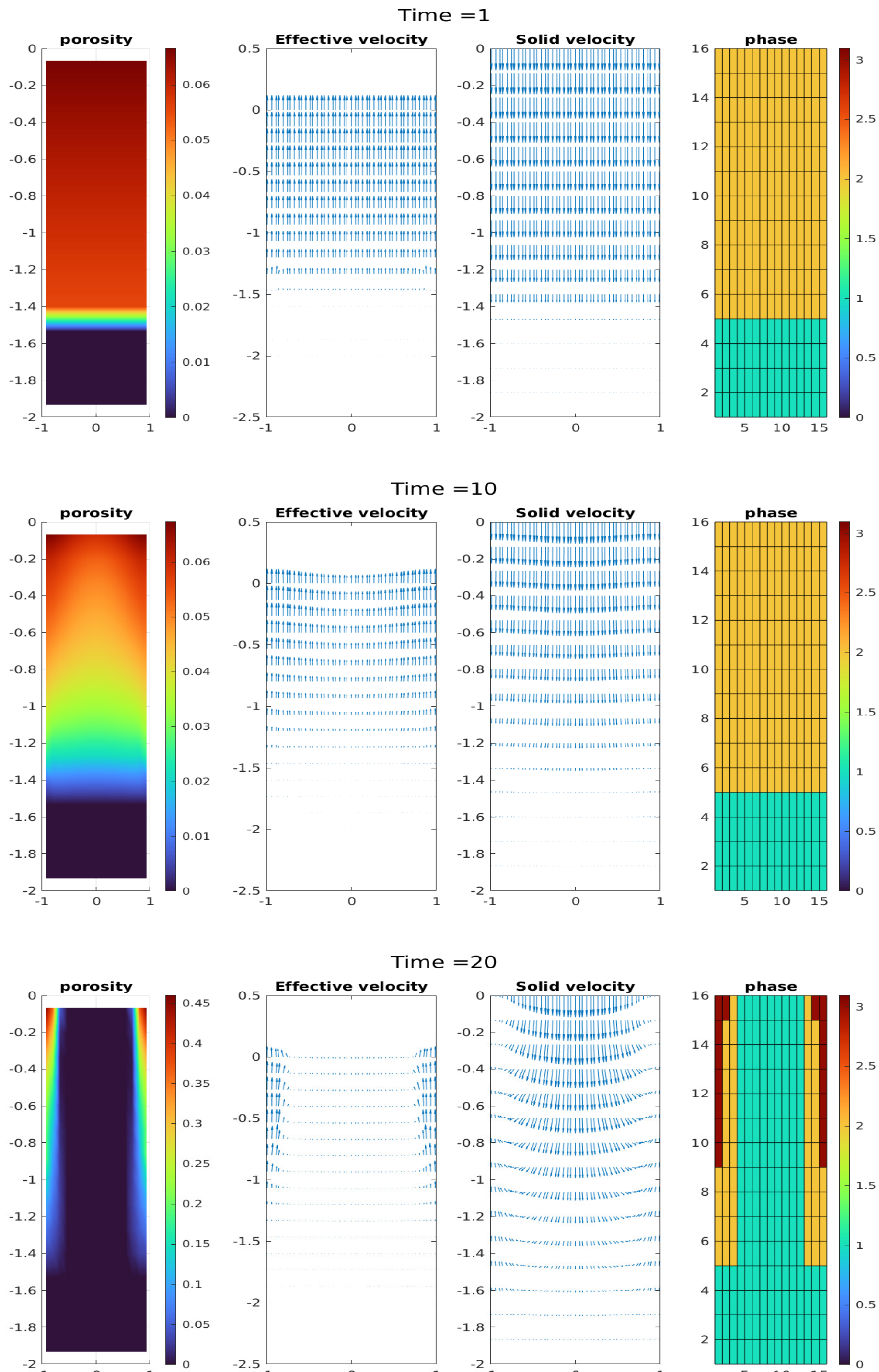


Figure 1.10: Full two-dimensional test case. Shown at time = 1, 10 and 20. We present porosity, effective velocity, solid velocity and phase split.

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